Decoder properties that hold with or without obstacles

Oct 13, 2026 · @jeffrey

Nine properties of the decoded travel time $T\theta(x, y) = D(Z\theta(x, y))$ hold for every obstacle configuration and can be guaranteed by the architecture alone, with no loss term. The reference design is a min-of-branches decoder $T = \min$ over k of $\sqrt{Q_k}$, each $Q_k \geq 0$ smooth. Properties that need a training loss or a check after training (Eikonal, Bellman, conditions along optimal paths, no false minima) are left out.

Metric properties

T is a distance on the free space, so these hold with obstacles. The triangle inequality is the exception for architecture: a general min of unrelated branches can break it, so it is built in only in the two special cases listed.

#	Property	Statement	How the architecture guarantees it
1	Nonnegative, zero exactly at the goal	$T \geq 0$, and $T(x, y) = 0$ only when $x = y$	Every $\sqrt{Q_k} \geq 0$; the head sets $Q_k(x, x) = 0$ for the branch through the goal and keeps every $Q_k > 0$ when $x \neq y$
2	Symmetry	$T(x, y) = T(y, x)$	Build each Q_k from a symmetric comparison of per-pose embeddings, e.g. $\|f(x) - f(y)\|^2$, or average a joint network over swapping x and y

#	Property	Statement	How the architecture guarantees it
3	Triangle inequality	$T(x, y) \leqslant T(x, z) + T(z, y)$	Only in two cases: (a) T is one distance between embeddings, $T = \|f(x) - f(y)\|$ in any norm, which has no ridges; (b) the branches are copies of one metric under a symmetry group, T = min over g of d(x, g·y), as with the ±q rotation branches
4	Speed-limit lower bound	$T \geqslant T_free = \sqrt{(\|\Delta t\|^2/s_t^2 + \theta^2/s_r^2)}$	Write each branch as T_free branch plus a residual $\delta_k \geqslant 0$ (e.g. softplus)

Structure at the goal

The goal is the one kink whose location and shape never depend on obstacles: it is always at x = y and always looks like the metric distance. This holds as long as the goal pose is not touching an obstacle.

#	Property	Statement	How the architecture guarantees it
5	Cone at the goal, in both endpoints	$T = \sqrt{Q}$ with Q smooth, $Q(x, x) = 0$ and $\nabla Q(x, x) = 0$	The square root in the decoder, on a branch whose Q_k vanishes quadratically at the goal
6	The cone has the metric's shape	Near the goal, T equals the metric distance to second order (formula below)	Use T_free plus a residual that vanishes quadratically at the goal, e.g. $\delta(x, y) = \|A(f(x) - f(y))\|^2$; T_free already has the exact metric shape, and δ adds nothing at first order

Property 6 in formulas, for a small body twist ξ at the goal p:

$$T\big(p\exp(\xi^\wedge),\,p\big)=\sqrt{g(\xi,\xi)}+O(|\xi|^2),\qquad \operatorname{Hess}_\xi\,Q\big(p\exp(\xi^\wedge),\,p\big)\Big|_{\xi=0}=2\operatorname{diag}\!\big(s_t^{-2}I_3,\ s_r^{-2}I_3\big)$$

Kinks

Away from the goal, T can only have ridges (peaks where two routes tie), and obstacles decide where those ridges are. These are the viscosity-solution conditions, which pick out the true travel time among the many functions that satisfy the Eikonal equation almost everywhere.

#	Property	Statement	How the architecture guarantees it
7	Peaks only	No V-shaped kink anywhere except the goal	A min of branches can only create peaks; avoid max, absolute value or ReLU on the output path except the goal's square root
8	Movable ridges	A ridge can form wherever two routes tie, at any pose-dependent location	The ridge is wherever two branches are equal, so its position is learned, not fixed
9	No cap on time	The decoder can output any travel time	Branch outputs are unbounded, so no bounded term forces detours into another latent

A min of branches rules out V-shaped kinks but not smooth dips, so ruling out false minima still needs the Bellman loss and is not listed here. The fixed (a, b) decoder fails 8 and 9 once obstacles are present: its ridge sits only at the half-turn and its rotation term is capped at π^2/s_r^2.